

Q1 - 25 July - Shift 1

A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force F , its length increases by 5 cm. Another wire of the same material of length $4L$ and radius $4r$ is pulled by a force $4F$ under same conditions. The increase in length of this wire is _____ cm.

*Space for your notes:***Q2 - 26 July - Shift 1**

In an experiment of determine the Young's modulus of wire of a length exactly 1m, the extension in the length of the wire is measured as 0.4mm with an uncertainty of ± 0.02 mm when a load of 1kg is applied. The diameter of the wire is measured as 0.4mm with an uncertainty of ± 0.01 mm. The error in the measurement of Young's modulus (ΔY) is found to be $x \times 10^{10} \text{ Nm}^{-2}$. The value of x is _____

Space for your notes:

[Take $g = 10 \text{ m/s}^2$]

Q3 - 26 July - Shift 2

Questions

MathonGo

The area of cross section of the rope used to lift a load by a crane is $2.5 \times 10^{-4} \text{ m}^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be :

(take $g = 10 \text{ ms}^{-2}$)

(A) $6.25 \times 10^{-4} \text{ m}^2$

(B) $10 \times 10^{-4} \text{ m}^2$

(C) $1 \times 10^{-4} \text{ m}^2$

(D) $1.67 \times 10^{-4} \text{ m}^2$

Space for your notes:

Q4 - 26 July - Shift 2

A uniform heavy rod of mass 20 kg. Cross sectional area 0.4 m^2 and length 20 m is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $x \times 10^{-9} \text{ m}$. The value of x is _____.
:(Given. Young's modulus $Y = 2 \times 10^{11} \text{ Nm}^{-2}$ and $g = 10 \text{ ms}^{-2}$)

Space for your notes:

Q5 - 27 July - Shift 1

A square aluminium (shear modulus is $25 \times 10^9 \text{ Nm}^{-2}$) slab of side 60 cm and thickness 15 cm is subjected to a shearing force (on its narrow face) of $18.0 \times 10^4 \text{ N}$. The lower edge is riveted to the floor. The displacement of the upper edge is _____ $\mu \text{ m}$.

Space for your notes:

Q6 - 27 July - Shift 2

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Questions

MathonGo

A steel wire of length 3.2 m ($Y_s = 2.0 \times 10^{11} \text{ Nm}^{-2}$) and a copper wire of length 4.4 m ($Y_c = 1.1 \times 10^{11} \text{ Nm}^{-2}$), both of radius 1.4 mm are connected end to end. When stretched by a load, the net elongation is found to be 1.4 mm. The load applied, in Newton, will be: (Given $\pi = \frac{22}{7}$)

- (A) 360 (B) 180 (C) 1080 (D) 154

Official Ans. by NTA (D)

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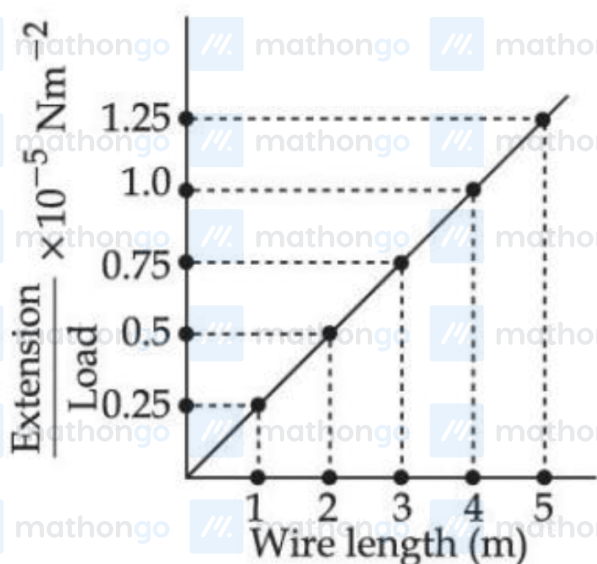
Q7 - 27 July - Shift 2

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Questions

MathonGo

In an experiment to determine the Young's modulus, steel wires of five different lengths (1, 2, 3, 4 and 5 m) but of same cross section (2 mm^2) were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11} \text{ Nm}^{-2}$, then the value of x is _____.

*Space for your notes:***Q8 - 28 July - Shift 1**

The force required to stretch a wire of cross-section 1 cm^2 to double its length will be :

(Given Young's modulus of the wire = $2 \times 10^{11} \text{ N/m}^2$)

- (A) $1 \times 10^7 \text{ N}$ (B) $1.5 \times 10^7 \text{ N}$
(C) $2 \times 10^7 \text{ N}$ (D) $2.5 \times 10^7 \text{ N}$

Space for your notes:

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Q9 - 28 July - Shift 2

A string of area of cross-section 4 mm^2 and length 0.5 is connected with a rigid body of mass 2 kg.

The body is rotated in a vertical circular path of radius 0.5 m. The body acquires a speed of 5 m/s at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is $\dots \times 10^{-5}$. (Use Young's modulus 10^{11} N/m^2 and $g = 10 \text{ m/s}^2$)

Space for your notes:

Q10 - 29 July - Shift 1

If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

- (A) Remains same
- (B) Become 8 times its initial value
- (C) Become $\frac{1}{4}$ of its initial value
- (D) Become 4 times its initial value

Space for your notes:

Q11 - 29 July - Shift 2

Questions

MathonGo

A metal wire of length 0.5 m and cross-sectional area 10^{-4} m^2 has breaking stress $5 \times 10^8 \text{ Nm}^{-2}$. A block of 10 kg is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be _____ ms^{-1} .

Space for your notes:

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Answer Key**Q1 (5)****Q2 (2)****Q3 (A)****Q4 (25)****Q5 (48)****Q6 (D)****Q7 (2)****Q8 (C)****Q9 (30)****Q10 (A)****Q11 (50)**

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Q1 (5)

$$\Delta \ell_1 = \frac{F\ell}{AY} = \frac{F\ell}{\pi r^2 Y} = 5\text{cm}$$

$$\Delta \ell_2 = \frac{4F4\ell}{\pi 16r^2 Y} = \frac{F\ell}{\pi r^2 Y} = 5\text{cm}$$

Q2 (2)

$$L = 1\text{m}$$

$$\Delta L = 0.4 \times 10^{-3} \text{ m}$$

$$m = 1\text{kg}$$

$$d = 0.4 \times 10^{-3} \text{ m}$$

$$\frac{F}{A} = Y \frac{\Delta L}{L}$$

$$Y = \frac{FL}{A\Delta L} = \frac{(mg).(1)}{\left(\frac{\pi d^2}{4}\right) 0.4 \times 10^{-3}}$$

$$\Rightarrow \frac{10 \times 4}{\pi (0.4 \times 10^{-3})^2 \times 0.4 \times 10^{-3}}$$

$$Y = \frac{40}{\pi (0.4 \times 10^{-3})^2 \times 0.4 \times 10^{-3}}$$

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$$Y = \frac{40 \times 7}{22 \times 64 \times 10^{-3} \times 10^{-9}}$$

$$Y = 0.199 \times 10^{-12} \text{ N/m}^2$$

$$\frac{\Delta Y}{Y} = \frac{\Delta F}{F} + \frac{\Delta L}{L} + \frac{\Delta A}{A} + \frac{\Delta(\Delta L)}{(\Delta L)}$$

$$= \frac{0.02}{0.4} + 2 \frac{\Delta d}{d} = \frac{0.2}{4} + 2 \times \frac{0.01}{0.4}$$

$$= \frac{0.1}{2} + \frac{0.1}{2} = 0.1$$

$$\Rightarrow \Delta Y = 0.1 \times Y$$

$$= 0.199 \times 10^{11} = 1.99 \times 10^{10}$$

Q3 (A)

Since breaking stress (Maximum lifting capacity)

is the property of material so it will remain same.

$$\text{breaking stress} = \frac{\text{Maximum lifting capacity}}{\text{Area of cross section of rope}}$$

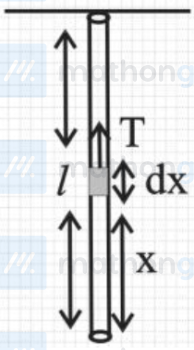
$$\frac{10}{2.5 \times 10^{-4}} = \frac{25}{A}$$

$$A = 625 \times 10^{-6}$$

$$= 6.25 \times 10^{-4} \text{ m}^2$$

Q4 (25)

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$$Y = \frac{T}{A} \frac{dx}{dy}$$

$$m = 20 \text{ kg}$$

$$A = 0.4 \text{ m}^2$$

$$l = 20 \text{ m}$$

let extension is dy in length dx

$$Y = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{\frac{T}{A}}{\frac{dx}{dy}} = \frac{T}{A} \cdot \frac{dy}{dx}$$

$$dy = \frac{T dx}{A Y}$$

$$\text{Tension at a distance } x \text{ from lower end} = \frac{mg}{\ell} x$$

$$\text{So, } \int_0^{\Delta l} dy = \int_0^{\ell} \frac{mg}{\ell} x \frac{dx}{A Y}$$

$$\Delta \ell = \frac{mg}{\ell A Y} \left[\frac{x^2}{2} \right]_0^{\ell}$$

$$\Delta \ell = \frac{mg \ell}{2 A Y}$$

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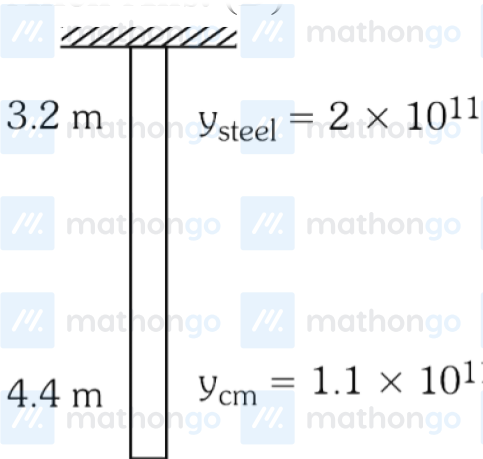
Q5 (48)

$$\frac{F}{A} = \eta \frac{x}{l} \Rightarrow \frac{F l}{A \eta} = x$$

$$\Rightarrow x = \frac{18 \times 10^4 \times 60 \times 10^{-2}}{60 \times 10^{-2} \times 15 \times 10^{-2} \times 25 \times 10^9}$$

$$= 48 \times 10^{-6} \text{ m} = 48 \mu \text{ m}$$

Q6 (D)



$$\Delta \ell_1 + \Delta \ell_2 = \Delta \ell$$

$$\frac{F \ell_1}{A_1 y_1} + \frac{F \ell_2}{A_2 y_2} = \Delta \ell$$

$$F = \frac{\Delta \ell}{\frac{\ell_1}{A_1 y_1} + \frac{\ell_2}{A_2 y_2}} = 1.54 \times 10^2 = 154$$

Q7 (2)

$$\text{Slope} = \frac{\Delta l / w}{L} = \frac{\Delta l / L}{w} = \frac{1}{YA}$$

$$\Rightarrow Y = \frac{1}{(\text{slope})A}$$

$$Y = \frac{1}{2 \times 10^{-6} (0.25 \times 10^{-5})}$$

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

Q8 (C)

$$F = \gamma A \frac{\Delta \ell}{\ell}$$

$$= 2 \times 10^{11} \times 10^{-4} \left(\frac{2\ell - \ell}{\ell} \right)$$

$$= 2 \times 10^7 \text{ N}$$

Q9 (30)



$$\text{Strain} = F/A Y$$

$$= \frac{mg + \frac{mv^2}{R}}{A Y}$$

$$= \frac{20 + \frac{2(5)^2}{0.5}}{3 \times 10^{-6} \times 10^{11}} = 30 \times 10^{-5}$$

Q10 (A)

Y depends on material of wire

Q11 (50)

$$T = \frac{mv^2}{\ell} = \frac{10 \times v^2}{0.5} = 20v^2$$

$$T_{\max} = \text{Breaking stress} \times \text{Area}$$
$$= 5 \times 10^8 \times 10^{-4} = 5 \times 10^4$$

$$20V^2 = 5 \times 10^4$$

$$V = \sqrt{\frac{1}{4} 10^4} = 50 \text{ m/s}$$